Measurement Technics
Project Report

Group 43 - KuDAQ

Authors Names

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Contents

1 Introduction

This report documents the experimental measurements and analysis conducted in the Measurement Technics course. The project focuses on the development and testing of the KuDAQ measurement system, which integrates various sensors for comprehensive data acquisition and analysis.

2 Measurements

2.1 Measurement 1: Orientation

2.1.1 Objective

The objective of the first measurement is to evaluate the impact of some environmental parameters on the movement of a standardized object caught in a jet flow. In order to do this, we will use the orientation of the object v.s. time, for a given set of parameters, during a fixed time interval. The jet flow is to be generated by a water pump in a water tank of approximately 5L capacity. The standardized object must be small enough to be free to move around the water tank, must not be physically linked to the acquisition system to remain fully independent to avoid potential biais, and must be heavy enough to be able to be caught in the jet flow, but not too heavy to be able to move around the water tank. The parameters that we were able to identify as potentially impacting the movement of the object are the following:

- The nozzle shape and size of the water pump
- The shape of the object
- Some other objects placed in the water tank
- The position of the water pump in the water tank
- The water level in the water tank

We eventually decided for this experiment to focus on the position of the water pump in the water tank, as it is a simple parameter to modify between tests, and that we expect to have a significant impact on the movement of the object. We planned to focus the measurement on the height of the water pump in the water tank, keeping the exact same horizontal position. Because the object will be floating on the water surface, we expect to find larger movement amplitude when the water pump is closer to the water surface, and smaller movement amplitude when the water pump is further from the water surface.

2.1.2 Equipment and Setup

Electrical Hardware

Mechanical Hardware Describe the mechanical hardware and physical setup:

- Mechanical mounting and fixtures
- Calibration reference devices
- Positioning and alignment tools
- Environmental control equipment

Software Architecture Describe the software architecture and computational framework:

- Data acquisition software and drivers
- Real-time processing framework
- Analysis tools (MATLAB/Python)
- Communication protocols

Acquisition Method Describe the data acquisition methodology:

- Sampling frequency and resolution
- Data storage format
- Synchronization method
- Trigger and timing specifications

2.1.3 Procedure

Describe the experimental procedure here. Include:

1. Sensor initialization and calibration
2. Test conditions and environmental parameters
3. Data collection methodology
4. Time duration and sampling frequency

2.1.4 Results

Present the measurement results and observations:

- Raw data from sensors
- Processed and filtered data
- Graphical representations and plots
- Statistical analysis

Data Analysis: Include relevant figures and tables here.

2.1.5 Discussion

Discuss the results in the context of:

- Expected behavior
- Observed deviations and anomalies
- Potential sources of error
- Accuracy and limitations

2.1.6 Conclusions

Summarize the key findings from this measurement and their implications.

3 Additional Measurements

3.1 Measurement 2: [Title to be determined]

Add content for additional measurements as needed, following the structure of Measurement 1.

4 Conclusion

Summarize the overall findings of the project, the effectiveness of the KuDAQ system, and recommendations for future work.

References

[1] Bosch Sensortec. BMI088 Datasheet.

[2] Madgwick, S. O. H., et al. An efficient orientation filter for inertial and inertial/magnetic sensor arrays.

A Raw Data

Include raw measurement data, extended tables, or additional plots here.

B Software and Configuration

Include code listings, configuration files, or technical documentation here.